Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/31/2024
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-7999
Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2024, per PI (Ed., 10/14/23) NOTE: End date changed to 9/30/2023, per J. Callas/JPL (Ed., 11/14/2022) NOTE: Changed JPL Task Plan number to 71-15884 per U. Israelsson/JPL (Ed., 5/27/2020) NOTE: End date changed to 9/30/2022, per PI (Ed., 2/27/2020) NOTE: End date changed to 9/30/2019, per PI (Ed., 11/17/17); changed to 10/1/2019 to accommodate final reporting (Ed., 10/4/18) NOTE: End date changed to 10/01/2017 (Ed., 11/14/16) NOTE: Extended for reflight to 9/30/2017 (per PI and U. Israelsson/JPL)-Ed., 5/12/14 NOTE: Extended to 3/31/2014 per PI (Ed., 10/21/13)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator: Fang Zhong, Ph.D is not currently with the project and has been removed from the list of Co-Investigators (Ed., 10/14/23).		

COI Name (Institution):	Beysens, Daniel Ph.D. (Commissariat a l'Energie Atomique at Grenoble and Ecole Supérieure de Physique et Chimie Industrielle) Garrabos, Yves Ph.D. (Centre National de la Recherche Scientifique (CNRS)-)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory) Lecoutre, Carole M.S. (Centre National de la Recherche Scientifique (CNRS)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory)
Grant/Contract No.:	JPL Task Plan number 71-15884 ; JPL Task Plan number 71-15890
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologists, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning Renormalization Group (RG) theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on "seemingly innocuous" assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient. The DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques)-EVO (Evolutionary) facility is an ideal platform to study the critical phenomena without gravity effects. Latest operations of the legacy DECLIC Alice-Like-Insert (ALI) have shown that the turbidity measurements could be performed very close to the critical point, taking advantages of the high-quality thermal and optical environment of the ALI insert and the microgravity condition of the International Space Station (ISS). The ALI cell was originally filled with sulfurhexafluoride (SF6) sample at slightly off-critical density for studying the boiling phenomena. The turbidity measurement at the off-critical density (~1.0% off from the critical density) was used to test and improve an Ising-like classical-to-critical crossover model. We also found that the minute deviation from the critical density value showed adverse effects for testing the true asymptotic nature of critical point phenomena. We proposed a re-flight of the Alice-Like-Insert (ALI-R) using a new cell with the same SF6 sample at much closer to the critical density and with an optimal geometry. The ALI-R will measure the effects of the non-zero value of the Green-Fisher exponent of the liquid-gas critical point. A measurement of the turbidity of the SF6 sample with a precision of 0.1% at 1 mK above the critical temperature (Tc=318.733 K) could test the theoretical predictions of Green-Fisher exponent.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work provides a unique opportunity to measure the experimental properties of a fluid near its liquid-gas critical point without gravitational effects, thereby refining theoretical equation-of-state models. Fluids in their critical and supercritical states exhibit unique properties. For instance, a fluid near the critical point and in a supercritical state shows very high compressibility. Using these fluids, for instance, in the storage of cryogenic propellants under reduced gravity conditions, raises fundamental questions regarding fluid dynamics, heat transfer, and interfacial phenomena. This experimental work will enhance understanding of the basic physics of fluids near the critical state.
Task Progress:	The ALI-R insert with the new sample cell was successfully assembled and tested on the ground using the DECLIC engineering model at CNES (Centre National d'Etudes Spatiales), France. The results from the ground-based tests were published. The ALI-R insert was launched for integration with the DECLIC facility aboard the International Space Station (ISS). During the facility operation of another insert experiment for studying solidification, the facility's He-Ne laser source failed, and the planned ALI-R experiment could not be performed. CNES successfully refurbished the DECLIC facility with a new laser, and NASA launched the refurbished "DECLIC-EVO" facility in late 2021 to perform the planned science investigations, including a continuation of the ALI-R experiment. The ALI-R insert was then successfully integrated with the DECLIC-EVO aboard the ISS, and we completed three ALI-R science sequences in 2022. The main science sequence involved performing light transmission measurements between 0.1 mK < Tc-T < 3 K. Two additional science operations originally planned were postponed due to a thermometer board failure and its associated computer booting anomaly. The CNES team modified the main computer boot sequence to bypass the troubled electronics board. In-orbit troubleshooting runs were successfully performed in June 2023, and the remaining two ALI-R science operations were completed in October, 2023 and May 2024. We analyzed the light transmission data to determine the turbidity of the fluid. With a remaining task by CNES to verify the thermometer calibration, the turbidity measurements are presented in the final report to NASA. We also developed a new parametric equation-of-state model, named the Crossover Master Model (CMM) to predict the thermodynamic properties and correlation length of any simple fluid near its gas-liquid critical point. This work is published in a peer-reviewed journal. The ALI-R insert returned to the ground in December 2024. The DECLIC-EVO facility is planned to be returned to the ground in May 2025.
Bibliography Type:	Description: (Last Updated: 01/02/2025)
Articles in Peer-reviewed Journals	Garrabos Y, Lecoutre C, Marre S, Hahn I. "Crossover Master Model of the equation-of-state for a simple fluid: Critical universality." Int J Thermophys. 2024 Jun;45(6):78. https://doi.org/10.1007/s10765-024-03359-7 , Jun-2024